

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE [PHASE-I & II]

TEST # 11

DATE : 02-12-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 11

Topics

DATE : 02 - 12 - 2018

PRE-MEDICAL NURTURE COURSE [PHASE-I & II] : [NEET]

PHYSICS : Centre of Mass, Collision, Rotational Motion, SHM

CHEMISTRY : Redox Reaction, Hydrogen, s-Block, p-Block

BIOLOGY : Transport In Plants + Mineral Nutrition, Digestive System, Respiratory System,
Circulatory System, Excretory System

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- | | |
|---|--|
| <p>1. A simple pendulum performs simple harmonic motion about $X = 0$ with an amplitude A and time period T. The speed of the pendulum at $X = \frac{A}{2}$ will be</p> <p>(1) $\frac{\pi A \sqrt{3}}{T}$ (2) $\frac{\pi A}{T}$</p> <p>(3) $\frac{\pi A \sqrt{3}}{2T}$ (4) $\frac{3\pi^2 A}{T}$</p> <p>2. A body of mass 5 gm is executing S.H.M. about a point with amplitude 10 cm. Its maximum velocity is 100 cm/sec. Its velocity will be 50 cm/sec at a distance</p> <p>(1) 5 (2) $5\sqrt{2}$</p> <p>(3) $5\sqrt{3}$ (4) $10\sqrt{2}$</p> <p>3. A particle executes S.H.M. with a period of 6 second and amplitude of 3 cm. Its maximum speed in cm/sec is</p> <p>(1) $\pi/2$ (2) π</p> <p>(3) 2π (4) 3π</p> <p>4. A body is executing S.H.M. When its displacement from the mean position is 4 cm and 5 cm, the corresponding velocity of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is</p> <p>(1) 2π sec (2) $\pi/2$ sec</p> <p>(3) π sec (4) $3\pi / 2$ sec</p> | <p>5. If a simple pendulum oscillates with an amplitude of 50 mm and time period of 2 sec, then its maximum velocity is</p> <p>(1) 0.10 m / s</p> <p>(2) 0.15 m / s</p> <p>(3) 0.8 m / s</p> <p>(4) 0.26 m / s</p> <p>6. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position the velocity of the particle is 10 cm/s. The distance of the particle from the mean position when its speed becomes 5 cm/s is</p> <p>(1) $\sqrt{3}$ cm (2) $\sqrt{5}$ cm</p> <p>(3) $2(\sqrt{3})$ cm (4) $2(\sqrt{5})$ cm</p> <p>7. The amplitude of a particle executing SHM is 4 cm. At the mean position the speed of the particle is 16 cm/sec. The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}$ cm/s, will be</p> <p>(1) $2\sqrt{3}$ cm (2) $\sqrt{3}$ cm</p> <p>(3) 1 cm (4) 2 cm</p> <p>8. The displacement equation of a particle is $x = 3\sin 2t + 4\cos 2t$. The amplitude and maximum velocity will be respectively</p> <p>(1) 5, 10 (2) 3, 2</p> <p>(3) 4, 2 (4) 3, 4</p> |
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SPACE FOR ROUGH WORK

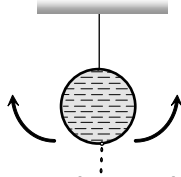
9. The instantaneous displacement of a simple pendulum oscillator is given by $x = A \cos\left(\omega t + \frac{\pi}{4}\right)$. Its speed will be maximum at time :-
- (1) $\frac{\pi}{4\omega}$ (2) $\frac{\pi}{2\omega}$ (3) $\frac{\pi}{\omega}$ (4) $\frac{2\pi}{\omega}$
10. A small body of mass 0.10 kg is executing S.H.M. of amplitude 1.0 m and period 0.20 sec. The maximum force acting on it is
- (1) 98.596 N (2) 985.96 N
(3) 100.2 N (4) 76.23 N
11. A body executing simple harmonic motion has a maximum acceleration equal to 24 metres/sec² and maximum velocity equal to 16 metres/sec. The amplitude of the simple harmonic motion is
- (1) $\frac{32}{3}$ metres (2) $\frac{3}{32}$ metres
(3) $\frac{1024}{9}$ metres (4) $\frac{64}{9}$ metres
12. A particle of mass 10 grams is executing simple harmonic motion with an amplitude of 0.5 m and periodic time of $(\pi/5)$ seconds. The maximum value of the force acting on the particle is
- (1) 25 N (2) 5 N
(3) 2.5 N (4) 0.5 N
13. A body is vibrating in simple harmonic motion with an amplitude of 0.06 m and frequency of 15 Hz. The velocity and acceleration of body is
- (1) 5.65 m/s and 5.32×10^2 m/s²
(2) 6.82 m/s and 7.62×10^2 m/s²
(3) 8.91 m/s and 8.21×10^2 m/s²
(4) 9.82 m/s and 9.03×10^2 m/s²
14. A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is
- (1) $\frac{1}{2\pi\sqrt{3}}$ (2) $2\pi\sqrt{3}$
(3) $\frac{2\pi}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2\pi}$
15. A particle is vibrating in a simple harmonic motion with an amplitude of 4 cm. At what displacement from the equilibrium position, is its energy half potential and half kinetic
- (1) 1 cm (2) $\sqrt{2}$ cm
(3) 3 cm (4) $2\sqrt{2}$ cm
16. The kinetic energy and potential energy of a particle executing simple harmonic motion will be equal, when displacement (amplitude = a) is
- (1) $\frac{a}{2}$ (2) $a\sqrt{2}$
(3) $\frac{a}{\sqrt{2}}$ (4) $\frac{a\sqrt{2}}{3}$

SPACE FOR ROUGH WORK

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| <p>17. The total energy of the body executing S.H.M. is E. Then the kinetic energy when the displacement is half of the amplitude, is</p> <p>(1) $\frac{E}{2}$ (2) $\frac{E}{4}$</p> <p>(3) $\frac{3E}{4}$ (4) $\frac{\sqrt{3}}{4}E$</p> <p>18. The potential energy of a particle executing S.H.M. is 2.5 J, when its displacement is half of amplitude. The total energy of the particle be</p> <p>(1) 18 J (2) 10 J</p> <p>(3) 12 J (4) 2.5 J</p> <p>19. The angular velocity and the amplitude of a simple pendulum is ω and a respectively. At a displacement X from the mean position if its kinetic energy is T and potential energy is V, then the ratio of T to V is</p> <p>(1) $X^2\omega^2/(a^2 - X^2\omega^2)$ (2) $X^2/(a^2 - X^2)$</p> <p>(3) $(a^2 - X^2\omega^2)$ (4) $(a^2 - X^2)/X^2$</p> <p>20. The total energy of a particle executing S.H.M. is 80 J. What is the potential energy when the particle is at a distance of $3/4$ of amplitude from the mean position</p> <p>(1) 60 J (2) 10 J</p> <p>(3) 40 J (4) 45 J</p> <p>21. A body executes simple harmonic motion. The potential energy (P.E.), the kinetic energy (K.E.) and total energy (T.E.) are measured as a function of displacement x. Which of the following statements is true</p> | <p>(1) P.E. is maximum when $x = 0$</p> <p>(2) K.E. is maximum when $x = 0$</p> <p>(3) T.E. is zero when $x = 0$</p> <p>(4) K.E. is maximum when x is maximum</p> <p>22. The total energy of a particle, executing simple harmonic motion is</p> <p>(1) $\propto x$</p> <p>(2) $\propto x^2$</p> <p>(3) Independent of x</p> <p>(4) $\propto x^{1/2}$</p> <p>23. The kinetic energy of a particle executing S.H.M. is 16 J when it is at its mean position. If the mass of the particle is 0.32 kg, then what is the maximum velocity of the particle</p> <p>(1) 5 m/s (2) 15 m/s</p> <p>(3) 10 m/s (4) 20 m/s</p> <p>24. The equation of motion of a particle is $\frac{d^2y}{dt^2} + Ky = 0$, where K is positive constant. The time period of the motion is given by</p> <p>(1) $\frac{2\pi}{K}$ (2) $2\pi K$</p> <p>(3) $\frac{2\pi}{\sqrt{K}}$ (4) $2\pi\sqrt{K}$</p> <p>25. The acceleration of a particle performing S.H.M. is 12 cm/sec^2 at a distance of 3 cm from the mean position. Its time period is :-</p> <p>(1) 0.5 sec (2) 1.0 sec</p> <p>(3) 2.0 sec (4) 3.14 sec</p> |
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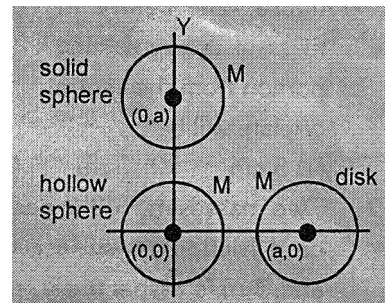
SPACE FOR ROUGH WORK

26. A simple pendulum is made of a body which is a hollow sphere containing mercury suspended by means of a wire. If a little mercury is drained off, the period of pendulum will



- (1) Remains unchanged
(2) First increase then decrease to initial value
(3) Decrease
(4) Become erratic
27. A pendulum suspended from the ceiling of a train has a period T , when the train is at rest. When the train is accelerating with a uniform acceleration a , the period of oscillation will
- (1) Increase
(2) Decrease
(3) Remain unaffected
(4) Become infinite
28. A simple pendulum is set up in a trolley which moves to the right with an acceleration a on a horizontal plane. Then the thread of the pendulum in the mean position makes an angle θ with the vertical :-
- (1) $\tan^{-1} \frac{a}{g}$ in the forward direction
(2) $\tan^{-1} \frac{a}{g}$ in the backward direction
(3) $\tan^{-1} \frac{g}{a}$ in the backward direction
(4) $\tan^{-1} \frac{g}{a}$ in the forward direction

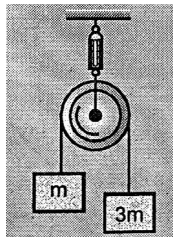
29. The time period of a second's pendulum is 2 sec. The spherical bob which is empty from inside has a mass of 50 gm. This is now replaced by another solid bob of same radius but having different mass of 100 gm. The new time period will be
- (1) 4 sec (2) 1 sec
(3) 2 sec (4) 8 sec
30. The coordinate of the centre of mass of a system as shown in figure:-



- (1) $\left(\frac{a}{3}, 0\right)$ (2) $\left(\frac{a}{2}, \frac{a}{2}\right)$
(3) $\left(\frac{a}{3}, \frac{a}{3}\right)$ (4) $\left(0, \frac{a}{3}\right)$
31. The centre of mass of a system of three particles of masses $1g$, $2g$ and $3g$ is taken as the origin of a coordinate system. The position vector of a fourth particle of mass $4g$ such that the centre of mass of the four particle system lies at the point $(1, 2, 3)$ is $\alpha(\hat{i} + 2\hat{j} + 3\hat{k})$, where α is a constant. The value of α is :-
- (1) $\frac{10}{3}$ (2) $\frac{5}{2}$ (3) $\frac{1}{2}$ (4) $\frac{2}{5}$

SPACE FOR ROUGH WORK

32. If the system is released, then the acceleration of the centre of mass of the system :-



- (1) $\frac{g}{4}$ (2) $\frac{g}{2}$
(3) g (4) $2g$
33. A heavy nucleus at rest breaks into two fragments which fly off with velocities 8 : 1. The ratio of radii of the fragments is :-
(1) 1 : 2 (2) 1 : 4
(3) 4 : 1 (4) 2 : 1
34. A bullet of mass m is fired from a gun of mass M . The recoiling gun compresses a spring of force constant k by a distance d . Then the velocity of the bullet is :-
(1) $kd\sqrt{M/m}$
(2) $\frac{d}{M}\sqrt{km}$
(3) $\frac{d}{m}\sqrt{kM}$
(4) $\frac{kM}{m}\sqrt{d}$
35. A sphere of mass m moving with a constant velocity collides with another stationary sphere of same mass. The ratio of velocities of two spheres after collision will be, if the co-efficient of restitution is e :-

(1) $\frac{1-e}{1+e}$ (2) $\frac{e-1}{e+1}$

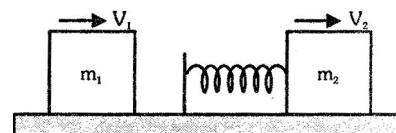
(3) $\frac{1+e}{1-e}$ (4) $\frac{e+1}{e-1}$

36. A particle falls from a height 'h' upon a fixed horizontal plane and rebounds. If V is the coefficient of restitution the total distance travelled before rebounding has stopped is :-

(1) $h\left(\frac{1+e^2}{1-e^2}\right)$ (2) $h\left(\frac{1-e^2}{1+e^2}\right)$

(3) $\frac{h}{2}\left(\frac{1-e^2}{1+e^2}\right)$ (4) $\frac{h}{2}\left(\frac{1+e^2}{1-e^2}\right)$

37. Two masses $m_1 = 2\text{kg}$ and $m_2 = 5\text{kg}$ are moving on a frictionless surface with velocities 10 m/s and 3 m/s respectively. m_2 is ahead of m_1 . An ideal spring of spring constant $k = 1120\text{ N/m}$ is attached on the back side of m_2 . The maximum compression of the spring will be :-



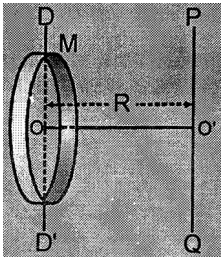
(1) 0.51 m (2) 0.062 m
(3) 0.25 m (4) 0.72 m

38. A solid sphere and a hollow sphere of the same mass have the same M.I. about their respective diameters. The ratio of their radii will be :-

(1) 1 : 2 (2) $\sqrt{3} : \sqrt{5}$
(3) $\sqrt{5} : \sqrt{3}$ (4) 4 : 5

SPACE FOR ROUGH WORK

39. The moment of inertia of a ring of mass M and radius R about PQ axis will be :-



- (1) MR^2 (2) $\frac{MR^2}{2}$
(3) $\frac{3}{2}MR^2$ (4) $2MR^2$
40. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4A_0$ in 4 seconds. The magnitude of this torque is :-
- (1) $\frac{3A_0}{4}$ (2) A_0
(3) $4A_0$ (4) $12A_0$
41. A round disc of moment of inertia I_2 about its axis perpendicular to its plane and passing through its centre is placed over another disc of moment of inertia I_1 rotating with an angular velocity ω about the same axis. The final angular velocity of the combination of discs is :-
- (1) ω (2) $\frac{I_1\omega}{I_1 + I_2}$
(3) $\frac{(I_1 + I_2)\omega}{I_1}$ (4) $\frac{I_2\omega}{I_1 + I_2}$

42. A small steel sphere of mass m is tied to a string of length r and is whirled in a horizontal circle with a uniform angular velocity 2ω . The string is suddenly pulled, so that radius of the circle is halved. The new angular velocity will be
- (1) 2ω (2) 4ω
(3) 6ω (4) 8ω
43. A ring is rolling without slipping. Its energy of translation is E . Its total kinetic energy will be :-
- (1) E (2) $2E$
(3) $3E$ (4) $4E$
44. If rotational kinetic energy is 50% of total kinetic energy then the body will be :-
- (1) ring
(2) cylinder
(3) hollow sphere
(4) solid sphere
45. Two rotating bodies A and B of masses m and $2m$ with moments of inertia I_A and I_B ($I_B > I_A$) have equal kinetic energy of rotation. If L_A and L_B be their angular momenta respectively, then :-
- (1) $L_B > L_A$
(2) $L_A > L_B$
(3) $L_A = \frac{L_B}{2}$
(4) $L_A = 2L_B$

SPACE FOR ROUGH WORK

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| <p>46. Which of the following is most abundant among group 13 elements in the earth's crust?</p> <p>(1) B (2) Al
(3) Ga (4) None of these.</p> <p>47. Boron does not form B^{3+} ions whereas Al forms Al^{3+} ions. This is because</p> <p>(1) the size of B atom is smaller than that of Al
(2) the sum of $IE_1 + IE_2 + IE_3$ of B is much higher than that of Al
(3) the sum of $IE_1 + IE_2 + IE_3$ of Al is much higher than that of B
(4) both (1) and (2)</p> <p>48. The hydroxide of which of the following element is amphoteric?</p> <p>(1) Al $(OH)_3$ (2) B $(OH)_3$
(3) In $(OH)_3$ (4) Tl $(OH)_3$</p> <p>49. The tendency of BF_3, BCl_3 and BBr_3 to behave as Lewis acid decreases in the sequence</p> <p>(1) $BF_3 > BCl_3 > BBr_3$
(2) $BCl_3 > BF_3 > BBr_3$
(3) $BBr_3 > BCl_3 > BF_3$
(4) $BBr_3 > BF_3 > BCl_3$</p> <p>50. H_3BO_3 is</p> <p>(1) monobasic and weak Lewis acid
(2) monobasic and weak Bronsted acid
(3) monobasic and strong Lewis acid
(4) tribasic and weak Bronsted acid</p> <p>51. Each B-H-B bridge in B_2H_6 is formed by the sharing of</p> <p>(1) 2 electrons (2) 4 electrons
(3) 1 electron (4) 3 electrons</p> | <p>52. Dry ice is</p> <p>(1) solid NH_3 (2) solid SO_2
(3) solid CO_2 (4) dry CO_2 gas.</p> <p>53. Glass reacts with HF to produce</p> <p>(1) Na_3AlF_6 (2) H_2SiF_6
(3) H_2SiO_3 (4) SiF_4</p> <p>54. BeF_2 is soluble in water, whereas the fluorides of other alkaline earth metals are insoluble because of :</p> <p>(1) ionic nature of BeF_2
(2) greater hydration energy of Be^{2+} ion as compared to lattice energy
(3) covalent nature of BeF_2
(4) none of these</p> <p>55. A metal [X] on heating in nitrogen gas gives [Y]. [Y] on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. [Y] is</p> <p>(1) $Mg(NO_3)_2$ (2) Mg_3N_2
(3) NaN_3 (4) MgO</p> <p>56. On the basis of structure of graphite which of the following is true for it.</p> <p>(1) It is a paramagnetic substance
(2) It behaves like metallic as well as semiconductor
(3) C-C bond length in it is more than the diamond
(4) It is thermodynamically more stable as well as more dense than diamond</p> |
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SPACE FOR ROUGH WORK

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| <p>57. Which one of the following statements is correct.</p> <p>(1) All boron trihalides are hydrolysed to boric acid.</p> <p>(2) Anhydrous aluminium chloride is an ionic compound</p> <p>(3) Aluminium halides make up the electron deficiency by bridging with halide</p> <p>(4) None of these</p> <p>58. Oxidation number of sulphur in S_8, S_2F_2 and H_2S are :-</p> <p>(1) +2, 0, +2</p> <p>(2) 0, +1, -2</p> <p>(3) -2, 0, +2</p> <p>(4) 0, +1, +2</p> <p>59. Amongst H_2O, H_2S, H_2Se and H_2Te the one with the highest boiling point is</p> <p>(1) H_2O because of hydrogen bonding</p> <p>(2) H_2Te because of higher molecular weight</p> <p>(3) H_2S because of hydrogen bonding</p> <p>(4) H_2Se because of lower molecular weight</p> <p>60. The compound that can work both as an oxidizing agent and a reducing agent is</p> <p>(1) $KMnO_4$ (2) H_2S</p> <p>(3) BaO_2 (4) H_2O_2</p> <p>61. Sodium reacts with water more vigorously than lithium because it</p> <p>(1) has higher atomic weight</p> <p>(2) is a metal</p> <p>(3) is more electropositive</p> <p>(4) more electronegative</p> | <p>62. The hydration energy of Mg^{2+} is greater than</p> <p>(1) Al^{+3} (2) Na^+</p> <p>(3) Be^{+2} (4) Mg^{+3}</p> <p>63. Which of the following metals do not impart colour to the flame?</p> <p>(1) Na (2) Mg</p> <p>(3) Ca (4) Sr</p> <p>64. The drying agent – which absorbs carbon dioxide and reacts violently with water is</p> <p>(1) sodium carbonate</p> <p>(2) alcohol</p> <p>(3) conc. H_2SO_4</p> <p>(4) calcium oxide</p> <p>65. Which of the following alkali metal halide has the lowest lattice energy?</p> <p>(1) LiF (2) NaCl</p> <p>(3) KBr (4) CsI</p> <p>66. On heating sodium metal in a current of dry ammonia, the compound formed is</p> <p>(1) sodium amide</p> <p>(2) sodium azide</p> <p>(3) sodium nitride</p> <p>(4) sodium hydride</p> <p>67. Borax structure contains</p> <p>(1) two BO_4 groups and two BO_3 groups</p> <p>(2) four BO_4 groups only</p> <p>(3) four BO_3 groups only</p> <p>(4) three BO_4 and one BO_3 groups</p> |
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SPACE FOR ROUGH WORK

68. The thermal stability of alkaline earth metal carbonates decreases in the order
 (1) $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$
 (2) $\text{BaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{CaCO}_3$
 (3) $\text{CaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{BaCO}_3$
 (4) $\text{MgCO}_3 > \text{CaCO}_3 > \text{SrCO}_3 > \text{BaCO}_3$
69. The correct order of ionic character is
 (1) $\text{BeCl}_2 < \text{MgCl}_2 < \text{CaCl}_2 < \text{BaCl}_2$
 (2) $\text{BeCl}_2 < \text{MgCl}_2 < \text{BaCl}_2 < \text{CaCl}_2$
 (3) $\text{BeCl}_2 < \text{BaCl}_2 < \text{MgCl}_2 < \text{CaCl}_2$
 (4) $\text{BeCl}_2 < \text{CaCl}_2 < \text{MgCl}_2 < \text{BaCl}_2$
70. The number of moles of KMnO_4 that will be needed to react completely with one mole of ferrous oxalate in acidic medium is
 (1) $\frac{3}{5}$ (2) $\frac{2}{5}$
 (3) $\frac{4}{5}$ (4) 1
71. The hourly energy requirement of an astronaut can be satisfied by the energy released when 34g of sucrose is burnt in his body. How many g of oxygen would be needed to be carried in space capsule to meet his requirement for one day?
 (1) 916.2gm
 (2) 91.62 gm
 (3) 8.162 gm
 (4) 9.162 gm
72. For the reaction

$$\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$$

 The correct coefficients of the reactants for the balanced reaction are
- | MnO_4^- | $\text{C}_2\text{O}_4^{2-}$ | H^+ |
|------------------|-----------------------------|--------------|
| (1) 2 | 5 | 16 |
| (2) 16 | 5 | 2 |
| (3) 5 | 16 | 2 |
| (4) 2 | 16 | 5 |
73. It takes 2.56×10^{-3} equivalents of KOH to neutralise 0.1254 g H_2XO_4 . The number of neutrons in X is
 (1) 16 (2) 8
 (3) 7 (4) 32
74. Equal volumes of 1 M each of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ are used to oxidise Fe(II) solution in acidic medium. The amount of Fe oxidised will be
 (1) more with KMnO_4
 (2) equal with both oxidising agents
 (3) more with $\text{K}_2\text{Cr}_2\text{O}_7$
 (4) cannot be determined
75. The number of moles of $\text{Cr}_2\text{O}_7^{2-}$ needed to oxidise 0.136 equivalents of N_2H_5^+ by the reaction:

$$\text{N}_2\text{H}_5^+ + \text{Cr}_2\text{O}_7^{2-} \longrightarrow \text{N}_2 + \text{Cr}^{3+} + \text{H}_2\text{O}$$

 (1) 0.136 (2) 0.272
 (3) 0.816 (4) 0.0227
76. The O – O – H bond angle in H_2O_2 in gas phase is
 (1) 106° (2) $109^\circ 28'$
 (3) 120° (4) 94.8°

SPACE FOR ROUGH WORK

77. Oxidation states of carbon atoms in diamond and graphite are :-
(1) +2, +4 (2) +4, +2
(3) -4, 4 (4) zero, zero
78. The maximum possible number of hydrogen bonds a water molecule can form in ice is
(1) 1 (2) 2 (3) 3 (4) 4
79. In the reaction, $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$, which choice has value twice that of equivalent mass of oxidising agent?
(1) 64 (2) 32 (3) 48 (4) 16
80. Which of the following carbide is known by the name hydrolith?
(1) MgC_2 (2) CaC_2
(3) SrC_2 (4) BeC_2
81. The correct order of solubility of sulphates of alkaline earth metals are
(1) $\text{BeSO}_4 > \text{MgSO}_4 > \text{SrSO}_4 > \text{CaSO}_4$
(2) $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4$
(3) $\text{BeSO}_4 > \text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4$
(4) $\text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4 > \text{BeSO}_4$
82. A fusion mixture is a mixture of
(1) $\text{Na}_2\text{CO}_3 + \text{MgCO}_3$
(2) $\text{Na}_2\text{CO}_3 + \text{K}_2\text{CO}_3$
(3) $\text{MgCO}_3 + \text{CaCO}_3$
(4) $\text{K}_2\text{CO}_3 + \text{CaCO}_3$
83. The compounds of alkaline earth metals have the following magnetic nature
(1) diamagnetic
(2) paramagnetic
(3) ferromagnetic
(4) anti ferromagnetic
84. Which one is used as an air purifier in space craft?
(1) quick lime
(2) slaked lime
(3) potassium superoxide
(4) anhydrous CaCl_2
85. The oxidation number of Cr in CrO_5 is
(1) +3 (2) +5 (3) +6 (4) 0
86. In which of the following reactions, there is no change in oxidation number for given atom
(1) $4\text{KClO}_3 \rightarrow 3\text{KClO}_4 + \text{KCl} \Rightarrow (\text{Cl})$
(2) $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 2\text{H}_2\text{O} + 3\text{S} \Rightarrow (\text{S})$
(3) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2 \Rightarrow (\text{S})$
(4) $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI} \Rightarrow (\text{I})$
87. In the reaction,
 $3\text{Cl}_2 + 6\text{NaOH} \rightarrow \text{NaClO}_3 + 5\text{NaCl} + 3\text{H}_2\text{O}$
the element which loses as well as gains electrons is
(1) Na (2) O
(3) Cl (4) none of three
88. In which one of the following compounds the oxidation number of oxygen is positive?
(1) H_2O_2 (2) Na_2O_2
(3) OF_2 (4) H_2O
89. Which of the following configuration is characteristic of group 13 (Group III A) elements?
(1) $ns^2 np^1$ (2) $(n-1)d^1 ns^2$
(3) $(n-1)d^{10} ns^2 np^2$ (4) $ns^2 np^3$
90. Boron halides behave as Lewis acids because they
(1) are covalent compounds
(2) are ionic compounds
(3) have only six electrons in the valence shell
(4) have a lone pair of electrons on the B atom.

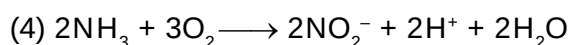
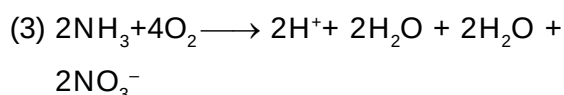
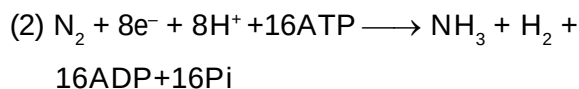
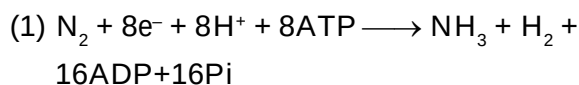
SPACE FOR ROUGH WORK

91. Which of the following techniques can researchers use to explore plant nutrient deficiencies?
 - (1) Hydroponics
 - (2) Sun exposure
 - (3) Crop rotation
 - (4) Hyperbaric chambers
92. Soil could be deficient in any of the following nutrients. If you had to supply one of them, which would be needed in the smallest amount?
 - (1) S
 - (2) P
 - (3) K
 - (4) Fe
93. In plants, a common symptom caused by deficiency of P, K, N and Mg is
 - (1) leaf tip hooking
 - (2) anthocyanin development
 - (3) necrosis/death of tissue
 - (4) poor formation of vascular tissue
94. Plants obtain Fe in the form of
 - (1) ferric ions
 - (2) ferrous ion
 - (3) either ferric or ferrous ions
 - (4) FeSO_4
95. Identify correct matching :-

Column-I		Column-II	
(1)	K	(A)	Stomatal opening
(2)	Mo	(B)	Constituent of cell membrane
(3)	P	(C)	Photolysis of water
(4)	Mn	(D)	Most free ion

96. Ammonifying bacteria in the soil
 - (1) converts ammonium to nitrate
 - (2) fixes nitrogen
 - (3) converts nitrogen in organic molecules into ammonium
 - (4) converts nitrogen to ammonia
97. The form of nitrogen from the soil that plant absorbs is
 - (1) NH_3
 - (2) N_2
 - (3) Nitrate
 - (4) Nitrite
98. Nodules that are actively fixing nitrogen are pink demonstrating the presence of
 - (1) Fe
 - (2) Chlorophyll
 - (3) Leg-haemoglobin
 - (4) Anthocyanin
99. Root nodules on plants of the legume family contain
 - (1) *Cyanobacteria*
 - (2) *Nitrococcus*
 - (3) *Rhizobium*
 - (4) *Nitrobacter*

100. Which one is the correct summary equation for the nitrogen fixation?



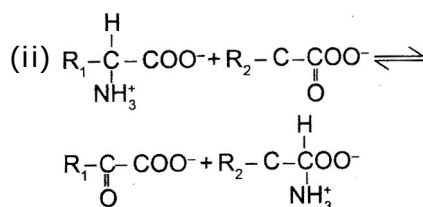
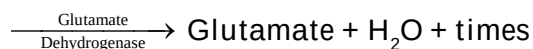
101. The process of conversion of $\text{NH}_4 \longrightarrow \text{NO}_2 \longrightarrow \text{NO}_3$ is called

- (1) Ammonification
- (2) Nitrification
- (3) N_2 fixation
- (4) Denitrification

102. Cell division in root nodules is promoted by _____ secreted by plants and _____ secreted by bacteria.

- (1) auxin, cytokinin
- (2) cytokinin, auxin
- (3) auxin, leg-haemoglobin
- (4) nitrogenase, leg-haemoglobin

103. NH_4^+ is used to synthesize amino acids in plants. " For it, there are two main ways:
(i) α -ketoglutaric acid + NH_4^+ + NADPH



- (1) Both (i) and (ii) are reductive amination
- (2) Both (i) and (ii) are transamination
- (3) (i) is transamination and (ii) is reductive amination
- (4) (i) is reductive amination and (ii) is transamination

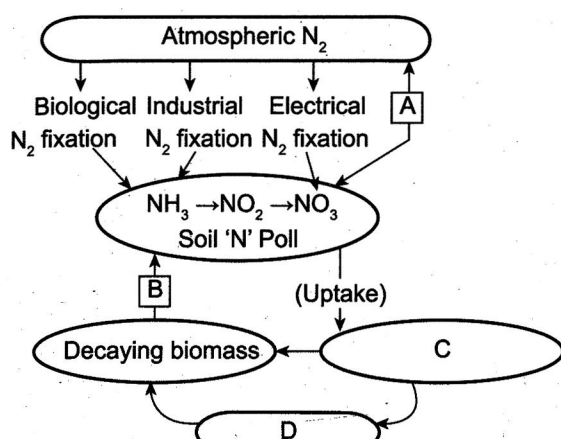
104. Conversion of $\text{NO}_3 \longrightarrow \text{NO}_2 \longrightarrow \text{NH}_4$ is called _____ and is catalysed by _____.

- (1) nitrate assimilation, nitrate and nitrite reductase
- (2) nitrification, nitrate and nitrate reductase
- (3) ammonification, glutamate dehydrogenase
- (4) denitrification, transaminase

105. Nitrite reductase enzyme is used to convert

- (1) nitrate into nitrite ion
- (2) nitrogen of atmosphere into ammonia
- (3) ammonia into nitrates
- (4) nitrite to ammonium ion

106. Study the cycle shown below and select the option which gives correct words for all the four blanks A, B, C and D.



	A	B	C	D
(1)	Nitrification	Ammonification	Animals	Plants
(2)	Denitrification	Ammonification	Plants	Animals
(3)	Nitrification	Denitrification	Animals	Plants
(4)	Denitrification	Nitrification	Plants	Animals

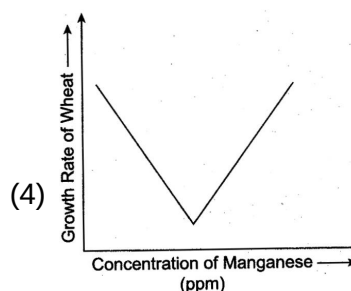
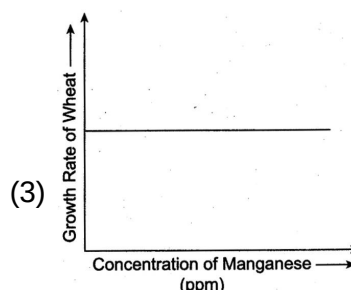
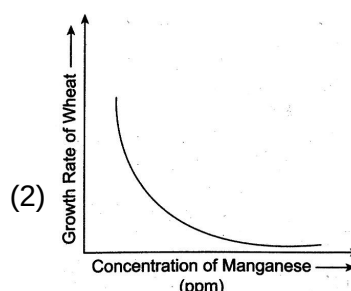
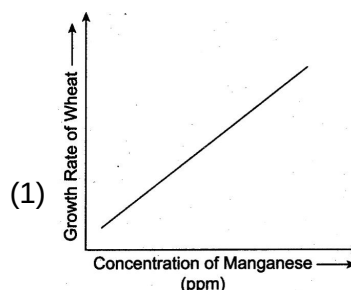
107. Nitrifying bacteria

- (1) reduces nitrates to free nitrogen
- (2) oxidize ammonia to nitrates
- (3) convert free nitrogen to nitrogen compounds
- (4) convert proteins into ammonia

108. The function of leg-haemoglobin in the root nodules of legumes is

- (1) expression of nif gene
- (2) inhibition of nitrogenase activity
- (3) oxygen removal
- (4) nodule differentiation

109. Mala and her classmates have been given a science assignment of studying the effect of concentration of micronutrients in the soil and the growth of crop plants. For this, the students decided to study the effect of manganese on the growth of wheat plants in the field. Which graph correctly illustrates the relationship between growth of plants and concentration of manganese in the soil?



110. Which one of the following is a wrong statement?

- (1) *Anabaena* and *Nostoc* are capable of fixing nitrogen in free living state also
- (2) Root nodule forming nitrogen fixers live as aerobes under free-living conditions
- (3) Phosphorus is a constituent of cell membranes, certain nucleic acids and cell proteins
- (4) *Nitrosomonas* and *Nitrobacter* are chemoautotrophs

111. Water potential and osmotic potential of pure water are

- (1) Zero and zero
- (2) Zero and 100
- (3) 100 and 100
- (4) 100 and zero

112. Which of the following criteria is not correct for facilitated diffusion?

- (1) Transport saturation
- (2) Highly selective
- (3) Requirement of special membrane proteins
- (4) Energy expenditure

113. Choose the incorrect match.

Property	Means of Transport
(1) Higher selective	Diffusion
(2) Transport saturation	Facilitated diffusion
(3) Uphill transport	Active transport
(4) Slow process	Simple diffusion

114. Gaseous movement with in the plant body is due to

- (1) Diffusion and sometime by active transport
- (2) Active transport only
- (3) Simple diffusion only
- (4) Facilitated diffusion and active transport

115. Simple diffusion differs from facilitated diffusion in

- (1) Showing uphill transport
- (2) Require ATP expenditure
- (3) Show transport saturation
- (4) No requirement of special Membrane proteins

116. DPD of a Fully turgid cell is

- (1) Zero
- (2) Equal to OP
- (3) Equal to TP
- (4) Negative

117. Maximum OP recorded in plant

- (1) *Tamarix*
- (2) *Atriplex*
- (3) *Prosopis*
- (4) *Mangifera*

118. The breaking of seed coat during seed germination is due to

- (1) Osmosis
- (2) Imbibition
- (3) Deplasmolysis
- (4) Verbalization

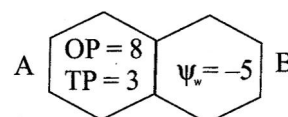
119. Two main components that determine water potential are

- (1) $\Psi_s + \Psi_p$
- (2) $\Psi_m + \Psi_g$
- (3) $\Psi_s + \Psi_g$
- (4) $\Psi_p + \Psi_g$

120. If a cell is placed in Hypertonic solution the protoplast starts shrinking and it first leaves the corner of cell at a stage is called

- (1) Incipient plasmolysis
- (2) Limiting plasmolysis
- (3) Evident plasmolysis
- (4) Deplasmolysis

121. Water will be direction of movement of water in two cells?

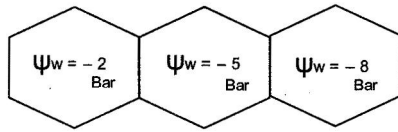


- (1) A to B
- (2) B to A
- (3) No net movement
- (4) Data deficient

122. During osmosis water passes through a semi-permeable membrane

- | | |
|-------------------------------|--------------------------|
| From | To |
| (1) High DPD | low DPD |
| (2) High solute concentration | low solute concentration |
| (3) High Ψ_w | low Ψ_w |
| (4) Hypertonic solution | Hypotonic solution |

123. In given diagram, the flow of water is possible from

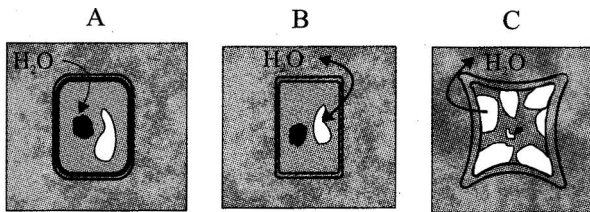


- (1) $A \rightarrow B \rightarrow C$ (2) $A \rightarrow B \leftarrow C$
(3) $A \leftarrow B \rightarrow C$ (4) $A \leftarrow B \leftarrow C$

124. In Flacid cell

- (1) $TP = 0$ (2) $DPD = 0$
(3) $\Psi_w = 0$ (4) $OP = 0$

125. Select the figure which represents the condition of cell when placed in Hypotonic solution.



- (1) A only (2) B only
(3) C only (4) (A) and (B) Both

126. When a cell is fully turgid, which of the following will be zero?

- (1) TP (2) WP
(3) Suction pressure (4) OP

127. Compared to 1 M sucrose solution the water potential (Ψ_w) of 1 M sodium chloride solution is

- (1) High (2) lower
(3) Same (4) None of these

128. In stomata, the cellulosic microfibrils of the cell wall of guard cell are oriented

- (1) Longitudinally
(2) Radially
(3) In all possible directions
(4) Radially as well as longitudinally

129. Lenticels and hydathodes are small pores with a common feature that is

- (1) Both are turgor operated valves
(2) They allow gaseous exchange
(3) No opening and closing mechanism
(4) All of these

130. An antitransparent is

- (1) Cobalt chloride
(2) Mercury
(3) Potassium
(4) Phenylmercuric acetate

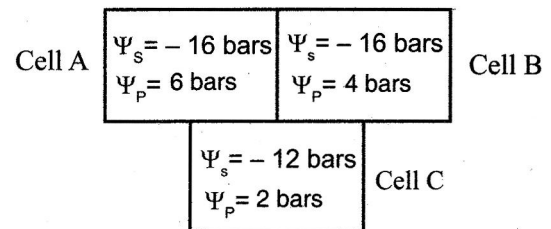
131. Phloem sap mainly contains

- (1) Water and Non-reducing sugar like sucrose
(2) Water and reducing sugar
(3) Water and amino acids
(4) water and organic Nitrogen

132. Sugar translocation in Flowering plants occurs by in the form of _____ and is a _____ process.

- (1) Sieve tubes, Starch, Bidirectional
(2) Sieve tubes, Sucrose, Bidirectional
(3) Companion cell, Glucose, Unidirectional
(4) Phloem parenchyma, Sucrose, Bidirectional

133. Find the direction of flow of water in the given system



- (1) From cell A to cell B
(2) From cell B to cell A
(3) No flow of water between cell A and cell C
(4) More than one option is correct

134. Statement-A : Transpiration pull model in the most acceptable to explain the transport of water.

Statement-B : The translocation in phloem is not explained by pressure flow hypothesis.

- (1) Only (A) is correct
(2) Only (B) is correct
(3) Both (A) and (B) are correct
(4) Both (A) and (B) are incorrect

135. Identify the statements are true or false.

- (A) Loading and unloading of sugar both utilises energy.
(B) Phloem sap includes water and sucrose only.
(C) Transport through phloem is essentially unidirectional
(D) Minerals undergo multidirectional transport

A	B	C	D	A	B	C	D
(1) T	F	F	T	(2) T	T	F	T
(3) T	T	F	F	(4) T	F	F	F

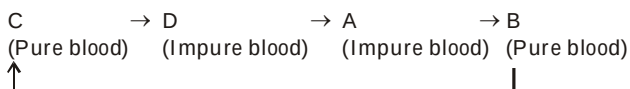
136. If myenteric plexus are completely cut off from the alimentary canal then :-

- (1) There will be increased digestion of carbohydrates.
- (2) There will increased digestion of proteins
- (3) There will increased digestion of fats
- (4) There will be no peristalsis movements.

137. Which of the following is the correct option w.r.t. pancreatic juice?

- (1) Trypsin and chymotrypsin are proteolytic endopeptidase
- (2) Trypsin can not coagulate milk
- (3) Maltase and invertase are present
- (4) Pancreatic lipase is called steapsin

138. Human heart has four chambers which are labelled as A, B, C & D. The direction of blood flow and its oxygen content is as shows below :-



Here the correct names of A, B, C & D will be :-

- (1) A – Right Atrium; B – Right Ventricle;
C – Left Atrium ; D – Left Ventricle
- (2) A – Right Ventricle; B – Left Atrium;
C – Left Ventricle ; D – Right Atrium
- (3) A – Left Atrium; B – Left Ventricle;
C – Right Atrium; D – Right Ventricle
- (4) A – Left Ventricle; B – Right Atrium;
C – Right Ventricle; D – Left Atrium

139. If six liters of deoxygenated blood enter the lungs via pulmonary artery in one minute time then the amount of oxygenated blood which will exit the lungs via pulmonary veins in a minute time will be :-

- (1) 6 litre
- (2) < 6 litre
- (3) > 6 litre
- (4) Data insufficient

140. Match the blood vessels of human heart listed under column-I with the functions given under column-II. Choose the answer which gives the correct combination of the alphabets of the two columns.

Column-I (Blood vessel)		Column-II (Function)	
(A)	Superior vena cava	p.	Carries deoxygenated blood to lungs
(B)	Inferior vena cava	q.	Carries oxygenated blood from lungs
(C)	Pulmonary artery	r.	Brings deoxygenated blood from lower parts of the body to the right atrium
(D)	Pulmonary vein	s.	Brings oxygenated blood to the left atrium
		t.	Brings deoxygenated blood from upper parts of the body into the right atrium

(1) A = t, B = p, C = r, D = q

(2) A = t, B = r, C = p, D = s

(3) A = s, B = t, C = r, D = p

(4) A = t, B = p, C = q, D = r

141. Regulation of kidney function of Juxtaglomerular Apparatus (JGA) involves certain steps given below. Arrange them in the correct order :-

- (a) Release of enzyme renin
- (b) Release of aldosterone from adrenal gland
- (c) Reabsorption of Na⁺ and water at DCT
- (d) Decrease in blood pressure and blood volume

(1) a, c, b, c, d

(2) b, c, a, d

(3) d, a, b, c

(4) d, a, b, c

142. Which of the following statements is incorrect?

- (1) Faecal accumulation in the rectum initiates a neural reflex causing an urge for its removal.
- (2) Irregular bowel movements cause constipation.
- (3) In diarrhoea absorption of food is increased
- (4) None of these

143. Which of the following is mismatched ?

- (1) Vitamin D - Rickets
- (2) Vitamin C - Scurvy
- (3) Vitamin K - Beriberi
- (4) Vitamin A - Xerophthalmia

144. Children suffering from Kwashiorkor have all of the following signs and symptoms, except

- (1) Swelling of legs due to water retention
- (2) Skin and hair changes
- (3) Pot belly due to fatty liver
- (4) Thin limbs and prominent ribs

145. Which of the following statement is incorrect?

- (1) Oxyntic cells release Castle's intrinsic factor (IF) that helps in absorption of cyanocobalamin.
- (2) Vit B₁₂ is also called extrinsic factor
- (3) Vit B₁₂ - IF complex gets absorbed in duodenum
- (4) Pernicious anaemia is an autoimmune disease in which antibodies are formed against IF

146. Which of the following are called protective principles of food?

- (1) Vitamin
- (2) Minerals
- (3) Water
- (4) All of these

147. Which of the following statements is not true :-

- (1) The partial pressure of O₂ in deoxygenated blood is 40 mm Hg.
- (2) The PO₂ in oxygenated blood is 95 mm Hg.
- (3) The PO₂ in alveolar air is 104 mm Hg.
- (4) The PCO₂ in deoxygenated is 95 mm Hg.

148. Match the items in column-I with column-II and choose the correct option :-

Column-I		Column-II	
A.	TV	I.	2500 to 3000 ml
B.	IRV	II.	1000 ml
C.	ERV	III.	500 ml
D.	RV	IV.	4800 ml
E.	VC	V.	1200 ml

A	B	C	D	E
(1) III	IV	II	I	V
(2) III	I	II	V	IV
(3) III	I	IV	V	IV
(4) IV	III	II	I	V

149. Read the following four statements (A-D)

- (i) Chest expand because air enters into the lungs
- (ii) Air enters into the lungs because chest expands
- (iii) The muscles of the diaphragm contracts because air enters in the lungs
- (iv) A healthy human breathes 12-16 times/minute

How many of the statement are wrong?

- (1) Four
- (2) Three
- (3) Two
- (4) One

150. Identify false statement :-

- (1) Dead space air is 150 ml
- (2) Air filled in bronchial tree is the dead space air
- (3) Bronchial tree collapses during expiration
- (4) Respiratory tree is inflated during inspiration

151. What is the state of heart in the interval after end of second heart sound and before beginning of first heart sound again :-

- (1) Ventricle remain in systole Auricles undergo first diastole and then systole
- (2) Ventricle remain in diastole Auricles undergo first diastole and then systole
- (3) Ventricles undergo first systole and then diastole Auricles remain in diastole
- (4) Ventricles remain in diastole Auricles remain in systole

152. Myoglobin binds _____ molecule of oxygen.

- (1) 4
- (2) 1
- (3) 3
- (4) 2

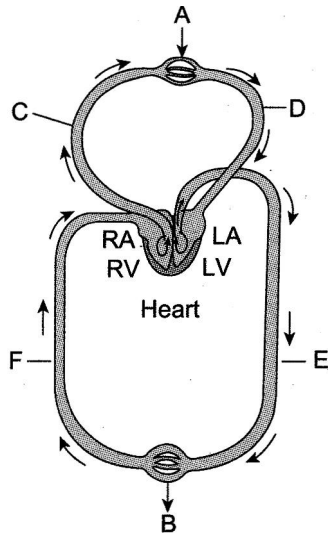
153. In methaemoglobin, iron is present in the form of
 (1) Ferrous ions
 (2) Ferric ions
 (3) Ferrous sulphate
 (4) Carboxyhaemoglobin
154. Which of the following statement correctly defines Bohr's effect?
 (1) Rise in P_{50} of haemoglobin with increase in CO_2 concentration
 (2) Fall in oxygen affinity of haemoglobin with increase in CO_2 concentration
 (3) Shift of oxygen dissociation curve of haemoglobin to right with increase in CO_2 concentration
 (4) All of these
155. Volume of oxygen present in one litre of deoxygenated blood under normal condition is
 (1) 50 ml (2) 150 ml
 (3) 200 ml (4) 25 ml
156. In comparison to adult haemoglobin, the oxygen-haemoglobin dissociation curve of foetal haemoglobin is situated towards
 (1) Left side
 (2) Right side
 (3) Either right or left
 (4) None of these
157. Yawning occurs due to
 (1) Excess of CO_2
 (2) Deficiency of oxygen
 (3) Excess of oxygen
 (4) Deficiency of CO_2
158. Find out the correct match :-

Column-I		Column-II	
(A)	Hepatic lobule	(I)	Base of villi
(B)	Crypts of leiberkuhn	(II)	Glisson's capsule
(C)	Sphincter of oddi	(III)	Gall bladder
(D)	Cystic duct	(IV)	Hepato - Pancreatic duct

- (1) A-II, B-IV, C-I, D-III
 (2) A-II, B-I, C-IV, D-III
 (3) A-I, B-II, C-III, D-IV
 (4) A-IV, B-III, C-II, D-I

159. Which of the following statements is correct?
 (1) Maximum water absorption occurs in small intestine
 (2) Small intestine is the major site of digestion and absorption of food.
 (3) Fatty acid and glycerol are absorbed by lacteals
 (4) All of the above
160. Chemoreceptors located in medulla respond to :-
 (1) High level of O_2 in blood
 (2) Low level of O_2 in blood
 (3) High level of CO_2 in blood
 (4) Low level of hydrogen ions in blood
161. Read the following points :-
 (i) Increase GFR
 (ii) Increase B.P.
 (iii) Decrease blood volume
 (iv) Increase aldosterone secretion
 (v) Vasoconstrictor
 (iv) Increases Na^+ excretion.
 How many points are correct about Angiotensin-II?
 (1) Two (2) Three
 (3) Four (4) Five
162. Volume of blood pumped out by each ventricle per minute is called :-
 (1) Stroke volume
 (2) Cardiac output
 (3) Cardiac index
 (4) Tidal volume
163. In ECG, P-wave represents
 (1) Atrial repolarization
 (2) Atrial depolarization
 (3) Ventricular depolarization
 (4) Ventricular repolarization
164. Pulse pressure is
 (1) Systolic pressure
 (2) Diastolic pressure
 (3) Pressure in vena cava
 (4) Difference between systolic and diastolic pressure

165. Which of the following is incorrect labelling w.r.t. the diagram given below?



- (1) D - Pulmonary vein
- (2) C - Pulmonary artery
- (3) E - Aorta
- (4) F - Carotid artery

166. If the heart sound recording and ECG recording are superimposed then the first heart sound would occur

- (1) At the P-wave
- (2) Just after P-wave
- (3) Just before QRS complex
- (4) Just after QRS complex

167. The greatest percentage of blood volume is found in

- (1) Veins
- (2) Arteries
- (3) Capillaries
- (4) Heart

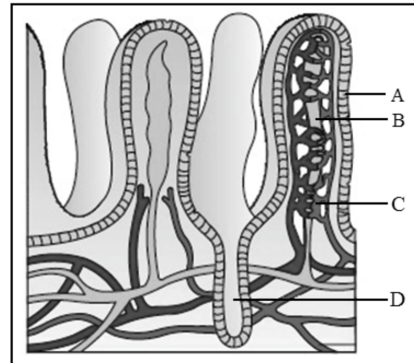
168. Death of a part of heart muscle due to cessation of blood supply to it is called

- (1) Heart failure
- (2) Heart attack
- (3) Heart block
- (4) Cardiac arrest

169. Aldosterone act mainly at the renal tubules and stimulate the rebsorption of :-

- (1) Sodium
- (2) Potassium
- (3) Both
- (4) Phosphate

170. The below diagram represents a section of small intestinal mucosa showing villi. Identify A, B, C and D.



- (1) A - Villi, B - Lacteul, C - Capillaries
D- Crypts
- (2) A - Lacteal, B - Capillaries, C - Villi,
D - Crypts
- (3) B - Villi, A - Lacteal, D - Crypts,
C - Capillaries
- (4) A - Crypts, B - Lacteal, C - Capillaries,
D - Villi

171. Consider the following four statements (A-D) and select the option which includes all correct :-

- (A) Henle's loop plays a significant role in the maintenance of high osmolarity of medullary interstitial fluid
- (B) Collecting duct plays a role in the maintenance of pH and ionic balance of blood by the selective secretion of H^+ and K^+ ions
- (C) Conditional reabsorption of Na^+ and water takes place in DCT
- (D) Nearly all of the essential nutrients and 70-80 percent of electrolytes and water are reabsorbed by DCT

- (1) Statements (A), (B), (C) only
- (2) Statements (A), (C), (D) only
- (3) Statements (B), (C) only
- (4) Statements (C), (D) only

172. Regulation of kidney function of Juxtaglomerular Apparatus (JGA) involves certain steps given below. Arrange them in the correct order :-

- Release of enzyme renin
- Release of aldosterone from adrenal gland
- Reabsorption of Na^+ and water at DCT
- Decrease in blood pressure and blood volume.

(e) Conversion of angiotensinogen to angiotensin II

- a, e, b, c, d
- b, c, a, e, d
- d, a, b, e, c
- d, a, e, b, c

173. Dialysis is done in the condition when person is suffering from :-

- Polyuria
- Haemolysis
- Pleurisy
- Uremia

174. Parasympathetic activity during micturition causes :-

- Contraction of detrusor muscle and contraction of internal urethral sphincter
- Contraction of detrusor muscle and relaxation of internal urethral sphincter
- Relaxation of detrusor muscle and relaxation of internal urethral sphincter
- Relaxation of detrusor muscle and contraction of Internal urethral sphincter

175. Maximum absorption of H_2O in nephron occurs from :-

- DCT
- Loop of Henle
- Ascending loop of Henle
- PCT

176. Choose the option showing correct matching of excretory disorders and its definition :-

Excretory disorder		Definition
(A)	Uremia	excess of urea in urine
(B)	Polyurea	more urea in blood
(C)	Cystitis	Presence of stone in kidney
(D)	Glomerulonephritis	Inflammation of Glomerulus

- A, B and D
- B, C and D
- Only D
- Only B

177. Which one is mainly responsible for absorption of Na^+ in the DCT part of Nephron :-

- Aldosterone
- ADH
- Angiotensin-II
- Atrial natriuretic factor (ANF)

178. Kidney stones are crystals of

- Sodium chloride
- Cholesterol
- Sodium bicarbonate
- Calcium oxalate

179. Lack of voluntary control over micturition is called

- Glycosuria
- Urinary incontinence
- Dysuria
- Uraemia

180. Accumulation of nitrogenous wastes in the blood is called

- Cystitis
- Anuria
- Azotemia
- Dysuria

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